



AQ10.1: Activity Questions 1 - Not Graded



This assignment will not be graded and is only for practice.

Level 1:

Consider a function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ defined as $f(x, y) = x^2 + 2xy^2 + 2x^2y + y^2$. If $\nabla f(1, 1) = (a, b)$, then find the value of $a - b$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

Consider the following function defined as

$$f_1(x, y) = \begin{cases} \frac{x}{y} & \text{if } y \neq 0 \\ 1 & \text{if } y = 0. \end{cases} \quad f_2(x, y) = \begin{cases} xy & \text{if } y \neq 0 \\ 0 & \text{if } y = 0. \end{cases}$$

$$f_2(x, y) = xy + x^2y^2 + x^2y$$

- **Statement 1:** f_1 has continuous gradient at the origin but f_2 does not have continuous gradient at the origin.
- **Statement 2:** f_2 has continuous gradient at the origin but f_1 does not have continuous gradient at the origin.
- **Statement 3:** $f_1(x, y)$ and $f_2(x, y)$ have continuous gradients at the origin.

Which of the above statements is correct?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

1 point

Consider a function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ defined as $f(x, y) = e^{(x^2 + xy + y^3)}$. $f(x, y) = e^{(x^2 + xy + y^3)}$.

Which of the following option(s) is (are) true?

☐

$$\nabla f = (e^{(x^2 + xy + y^3)}, e^{(x^2 + xy + y^3)}) \quad \nabla f = (e^{(x^2 + xy + y^3)}, e^{(x^2 + xy + y^3)})$$

☐

$$\nabla f = (2x + y, 3y^2 + x) \quad \nabla f = (2x + y, 3y^2 + x)$$

☐

$$\nabla f = (2xe^{(x^2 + xy + y^3)}, 3y^2 e^{(x^2 + xy + y^3)}) \quad \nabla f = (2xe^{(x^2 + xy + y^3)}, 3y^2 e^{(x^2 + xy + y^3)})$$

☐

$f(x, y)$ increases most rapidly in the direction of $\frac{1}{5}(3, 4)$ at the point $(1, 1)$.

☐

$f(x, y)$ decreases most rapidly in the direction of $\frac{1}{\sqrt{17}}(-1, -4)$ at the point $(1, -1)$.

☐

$\sqrt{17}(-1, -4)$ at the point $(1, -1)$.



No, the answer is incorrect.

Score: 0

Accepted Answers:

$f(x, y)$ increases most rapidly in the direction of $\frac{1}{5}(3, 4)$ at the point $(1, 1)$.

$f(x, y)$ decreases most rapidly in the direction of $\frac{1}{\sqrt{17}}(-1, -4)$ at the point $(1, 1)$.

$\sqrt{17}$
 $(-1, -4)$ at the point $(1, 1)$.

Level 2:

With a particular frame of reference (in $\mathbb{R}^2$), a corner of a hot rectangular iron plate with some finite length and breadth is placed at the origin $(0, 0)$. The temperature of the plate at a point (x, y) is given by the function:

$$T(x, y) = xy^2 + x^2y$$

Use this information to answer the following questions

1 point

From the point $(1, 3)$, in which direction the rate of change in temperature is the minimum?

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$\frac{1}{\sqrt{274}}(15, 7)$

☐

$\frac{1}{\sqrt{82}}(-9, -1)$

☐

$\frac{1}{\sqrt{82}}(9, 1)$

☐

$\frac{1}{\sqrt{82}}(-15, -7)$

☐

$\frac{1}{\sqrt{82}}(15, 7)$

☐

$\frac{1}{\sqrt{274}}(-15, -7)$

☐

$\frac{1}{\sqrt{274}}(15, 7)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\frac{1}{\sqrt{274}}(15, 7)$

☐

$\frac{1}{\sqrt{82}}(-9, -1)$

1 point

From the point $(1, 3)$, in which direction the rate of change in temperature is the maximum?

☐

$\frac{1}{\sqrt{274}}(15, 7)$

☐

$\frac{1}{\sqrt{82}}(-9, -1)$

☐

$$\frac{1}{\sqrt{82}} \quad 82$$

☐

$$1(9, 1)$$

☐

$$\frac{1}{\sqrt{82}} \quad 82$$

☐

$$1(-9, -1)$$

☐

$$\frac{1}{\sqrt{274}} \quad 274$$

☐

$$1(15, 7)$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\frac{1}{\sqrt{274}} \quad 274$$

☐

$$1(15, 7)$$

1 point

From a point (1,3), in which direction the rate of change in temperature of the iron plate is zero?

☐

$$\frac{1}{\sqrt{274}} \quad 274$$

☐

$$1(-7, 15)$$

☐

$$\frac{1}{\sqrt{82}} \quad 82$$

☐

$$1(1, -9)$$

☐

$$\frac{1}{\sqrt{82}} \quad 82$$

☐

$$1(-1, 9)$$

☐

$$\frac{1}{\sqrt{274}} \quad 274$$

☐

$$1(7, -15)$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\frac{1}{\sqrt{274}} \quad 274$$

☐

$$1(-7, 15)$$

$$\frac{1}{\sqrt{274}} \quad 274$$

☐

$$1(7, -15)$$

1 point

Consider a function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ defined as $f(x, y, z) = 2e^{2x} \sin yz$

$f(x, y, z) = 2e^{2x} \sin yz$. At the point $(0, \frac{\pi}{2}, 12\pi, 1)$, which are the directions that have the directional

derivative to be 0 ?

☐

$$\frac{1}{\sqrt{2}} \quad 2$$

☐

$$\sqrt{1} \quad (0,1,1)$$

☐

$$\frac{1}{\sqrt{2}} \quad 2$$

☐

$$\sqrt{1} \quad (0,1,-1)$$

☐

$$\frac{1}{\sqrt{3}} \quad 3$$

☐

$$\sqrt{1} \quad (1,1,1)$$

☐

$$\frac{1}{2021\sqrt{2}} \quad 2021 \quad 2$$

☐

$$\sqrt{1} \quad (0,2021,-2021)$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\frac{1}{\sqrt{2}} \quad 2$$

☐

$$\sqrt{1} \quad (0,1,1)$$

$$\frac{1}{\sqrt{2}} \quad 2$$

☐

$$\sqrt{1} \quad (0,1,-1)$$

$$\frac{1}{2021\sqrt{2}} \quad 2021 \quad 2$$

☐

$$\sqrt{1} \quad (0,2021,-2021)$$

1 point

Consider a function $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ defined as $f(x, y, z) = \alpha x + \beta x y^2 + \gamma y z^2$

$f(x, y, z) = \alpha x + \beta x y^2 + \gamma y z^2$. If the maximum value of the directional derivative of $f(x, y, z)$ at $(1, 1, 1)$ is in the direction parallel to x -axis and has magnitude 5, then

☐

$$f(x, y, z) = 5x + x^2 \quad f(x, y, z) = 5x + x^2$$

☐

$$f(x, y, z) = 2x + x y^2 \quad f(x, y, z) = 2x + x y^2$$

☐

$$f(x, y, z) = 5x \quad f(x, y, z) = 5x$$

☐

$$f(x, y, z) = x y^2 + 2 y z^2 \quad f(x, y, z) = x y^2 + 2 y z^2$$

☐

$$f(x, y, z) = 2x \quad f(x, y, z) = 2x$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$f(x, y, z) = 5x \quad f(x, y, z) = 5x$$

Check Answers